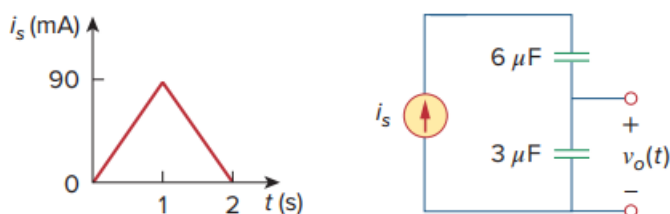


Problem

Assuming that the capacitors are initially uncharged, find $v_o(t)$ in the circuit of Fig. 6.62.

Fig. 6.62:



Step-by-step solution

Step 1 of 3

Refer to current waveform and circuit diagram in Figure 6.62 in the textbook.

Determine current for the interval $0 < t < 1$.

$$i_s(t) - 0 = \left(\frac{90 \times 10^{-3} - 0}{1 - 0} \right) (t - 0)$$

$$i_s(t) = 90t \text{ mA}$$

Determine current for the interval $1 < t < 2$.

$$i_s(t) - 90 \times 10^{-3} = \left(\frac{0 - 90 \times 10^{-3}}{2 - 1} \right) (t - 1)$$

$$i_s(t) - 90 \times 10^{-3} = -90 \times 10^{-3} t + 90 \times 10^{-3}$$

$$i_s(t) = (180 - 90t) \text{ mA}$$

The expression for $i_s(t)$ is,

$$i_s(t) = \begin{cases} 90t \text{ mA}, & \text{for } 0 < t < 1 \\ (180 - 90t) \text{ mA}, & \text{for } 1 < t < 2 \end{cases}$$

Step 2 of 3

Calculate the value of the voltage $v_o(t)$ across the $3 \mu\text{F}$ capacitor for the interval $0 < t < 1 \text{ s}$.

$$\begin{aligned} v_o(t) &= \frac{1}{C} \int_0^t i_s(t) dt + v_o(0) \\ &= \frac{1}{3 \times 10^{-6}} \int_0^t (90 \times 10^{-3}) t dt + 0 \\ &= \left(\frac{90 \times 10^{-3}}{3 \times 10^{-6}} \right) \frac{t^2}{2} \bigg|_0^t \\ &= 15t^2 \text{ kV} \end{aligned}$$

Calculate the voltage at $t = 1 \text{ s}$.

$$\begin{aligned} v_o(1 \text{ s}) &= 15000(1)^2 \text{ V} \\ &= 15 \text{ kV} \end{aligned}$$

Step 3 of 3

Calculate the value of the voltage $v_o(t)$ across the $3\text{ }\mu\text{F}$ capacitor for the interval $1 < t < 2\text{ s}$.

$$\begin{aligned}v_o(t) &= \frac{1}{C} \int_1^t i_s(t) dt + v_o(1\text{ s}) \\&= \frac{1}{3 \times 10^{-6}} \int_1^t (180 \times 10^{-3} - 90 \times 10^{-3} t) dt + 15 \times 10^3 \\&= \frac{90 \times 10^{-3}}{3 \times 10^{-6}} \left(2t - \frac{t^2}{2} \right)_1^t + 15 \times 10^3 \\&= (30000) \left(2t - \frac{t^2}{2} - 2 + \frac{1}{2} \right) + 15 \times 10^3 \\&= 60 \times 10^3 t - 15 \times 10^3 t^2 - 45 \times 10^3 + 15 \times 10^3 \\&= -15t^2 + 60t - 30\text{ kV}\end{aligned}$$

Therefore, the voltage $v_o(t)$ across the $3\text{ }\mu\text{F}$ capacitor is

$$\boxed{\begin{cases} 15t^2\text{ kV}, & \text{for } 0 < t < 1\text{ s} \\ (-15t^2 + 60t - 30)\text{ kV}, & \text{for } 1 < t < 2\text{ s} \end{cases}}.$$